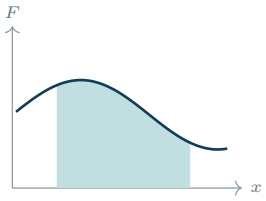
MECHANICS AND FIELDS ON ONE PAGE

Ideal forces, measurable integrals

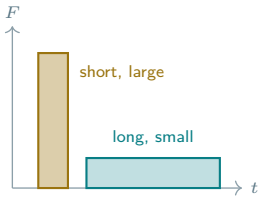
Work, impulse, pressure and flux are what a force or field gives when it is read over a distance, a time or an area.

1 / THE RULE AND THE PICTURE

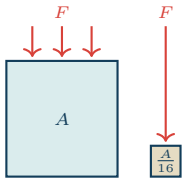
A force read over a distance gives work, over a time gives impulse, over an area gives pressure



work
$W = \int F\,dx$



impulse, equal areas
$J = \int F\,dt$



pressure
$p = F/A$, then $16p$

A force F or a field E is defined at a point and an instant, and an instrument can only read it over a window R, a stretch of distance, a span of time or a patch of area. The reading is the integral of the local value over the window, and the local value is recovered as the limit of the reading per unit window:

$$Q(R) = \int_R q, \qquad q = \lim_{R \to \text{point}} \frac{Q(R)}{|R|}$$

2 / THE SAME FORCE AND FIELD OVER DIFFERENT WINDOWS

Every measurable quantity is an integral or a ratio over a finite region

ideal local object	read over	measurable quantity	unit	worked example
force F	a distance	work $W = \int F \cdot dx$	N m = J	spring $k = 200$ N/m stretched 0.1 m: $\frac{1}{2}kx^2 = 1$ J
force F	a time interval	impulse $J = \int F\,dt$	N s	50 N for 0.2 s: $J = 10$ N s
force F	an area	pressure $p = F/A$	N/m^2 = Pa	600 N on 200 cm^2: 30 kPa, on 1 cm^2: 6 MPa
field E	an area	flux $\Phi = \int E \cdot dA$	V m	100 V/m through 0.5 m^2, $60°$ from the normal: 25 V m
irradiance I	an area	power $P = \int I\,dA$	W	sunlight in space, 1361 W/m^2 on 2 m^2: 2.72 kW
power P	a time interval	energy $\int P\,dt$	J	2.72 kW for one hour: 9.8 MJ

3 / TWO INTEGRALS OF ONE LAW

Newton's second law $F = \frac{dp}{dt}$ read over a time and over a distance

Integrating $F = \frac{dp}{dt}$ over a time interval gives the impulse $J = \int F\,dt = \Delta p$, and multiplying by $v = \frac{dx}{dt}$ and integrating over the distance gives the work $W = \int F\,dx = \Delta K$ with $K = \frac{p^2}{2m}$. For 50 N on a 2 kg mass from rest, after 0.2 s the impulse is 10 N s, the speed 5 m/s, the distance 0.5 m and the work 25 J, which equals $K = \frac{1}{2} \cdot 2 \cdot 5^2$. Doubling the time doubles the impulse to 20 N s and the speed to 10 m/s, and quadruples the distance and the work to 2 m and 100 J. Both identities hold for any force: for $F = 40 + 30\sin 5t$ newtons on 2 kg over 0.5 s, $J = 30.81$ N s and $W = 237.3$ J $= \frac{J^2}{2m}$.

4 / WHY THE LOCAL VALUES ARE IDEAL

A reading depends on its window, and the integral does not

A force sensor that samples over a window Δ returns the mean $\frac{1}{\Delta}\int F\,dt = \frac{J_\Delta}{\Delta}$. A tennis ball of 0.058 kg that reverses from 30 m/s to -30 m/s receives the impulse 3.48 N s, so a window matching a 5 ms contact reads 696 N, a window of 50 ms reads 69.6 N and one of 0.1 s reads 34.8 N, while the impulse is the same in all three. The force of the collision, a spike with area J, is an idealisation that no window reaches. Pressure has the same limit, since below the mean free path of air, about $0.07\,\mu$m, force arrives as separate molecular impacts and no pressure can be read. A window of width Δ also removes any oscillation of which a whole number of periods fits inside it, so detail finer than the window is invisible, which is the relation between a distance d and the spatial frequency $\frac{1}{d}$.

5 / WHAT SURVIVES A CHANGE OF WINDOW

Integrals add over windows, and averages need a window large enough to smooth the fluctuations

Work, impulse, flux and energy add over disjoint windows, as in $W_{0\to2} = W_{0\to1} + W_{1\to2}$ (both 12 J for $F = 3x^2 + 2$), so refining a window only subdivides them. Pressure, irradiance and a mean force are ratios of such quantities to the window and shift with it until the window holds enough events: a count of N photons or molecules fluctuates by $\frac{1}{\sqrt{N}}$, which is 1% for $N = 10^4$. Sunlight of 1361 W/m^2, treated as 550 nm light, delivers about 3.8×10^3 photons to $1\,\mu$m^2 in $1\,\mu$s, a fluctuation of 1.6% in the irradiance read there.

6 / THE DIVISION OF LABOUR

Local objects carry the differential laws and windowed quantities carry the measurements

The local objects F, E and I appear in the differential laws, and the windowed quantities W, J, Φ and P are what instruments return. The integral theorems connect the two by saying that what is read over an interval, a path or a closed surface equals a change or a content inside: $J = \Delta p$ over a time, $W = \Delta K$ over a path, and Gauss's law $\Phi = \frac{Q}{\varepsilon_0}$ over a closed surface, where 1 nC enclosed gives 113 V m through any surface around it.

ASIDE / TRY THESE

Show that the tennis ball reads 34.8 N over 0.1 s, and find the area of window that holds 10^4 photons of this sunlight in $1\,\mu$s, which is about $2.7\,\mu$m^2.

LINKS TO OTHER SHEETS

The Fourier sheet treats a window as a filter, and the Pascal sheet counts how discrete outcomes spread into a distribution.

Convention. F force, E field, I irradiance, p momentum or pressure by context, K kinetic energy; all worked values were checked by computation.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames